

Gate (II.11), phase V4:

Fixed- t convergence, partition functions and the derived gap, and the assembly: the gate closes

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Abstract

This document executes the final phase of the thermal two-torus route: Statements A6, A7, A8 of the scoping note are proved, and the gate

$$\lim_{N \rightarrow \infty} \omega_{N,L,\lambda,1}(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))) \quad (\text{every fixed } M, \xi) \quad (\text{II.11})$$

is closed after the 16 August repair of V2's four-edge and shifted-symbol arguments. A6 (fixed- t convergence of the thermal functionals) is assembled from the two mean-shift representations (V1 for the lattice, V3 for the continuum) compared on the V2 coupling space; the two remaining quantitative pieces — convergence of the action prefactors and of the marked-slice source phases — are proved here from the V1 thermal weight bounds and the V2 filter-tail and symbol estimates. A7 (partition functions) is proved in the normal-ordered normalization (a recorded amendment to the scoping display, whose un-normalized free products vanish in the limit); the scoping note's Laplace-transfer lemma then delivers eigenvalue convergence $\tilde{E}_j(N) \rightarrow \mu_j$, the derived eventual gap $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$, and $D_N(t) \rightarrow D_\infty(t)$. A8 verifies the hypotheses of the scoping note's interchange lemma and invokes the proved Weyl criterion and identification corollary: the full-sequence, identified, fixed-coarse theorem follows, with the $\mathbb{Z}_2$ symmetry carried along. A claim ledger, the erratum register for upstream documents, and the remaining phase-closure step (the adversarial pass) conclude the document.

Contents

1	Setup, deliverables, imports	2
1.1	Objects	2
1.2	Deliverables	2
1.3	Imports	2
2	The two representations on the coupling space	6
3	Two comparison lemmas	6
4	A6: fixed-t convergence of the thermal functionals	9
5	A7: partition functions, Laplace transfer, the derived gap	9
6	A8: assembly — the gate closes	10
7	Acceptance, ledger, and phase closure	12

1 Setup, deliverables, imports

1.1 Objects

Conventions of the model sheet and the phase documents V1–V3 are used unchanged. Fix $L, m > 0$, $\lambda \geq 0$, the coarse scale M and a coarse Weyl symbol $\xi = \varepsilon_M q + ip \in h_{M,L}$; $N = M + r$ runs over the fine scales, $\varepsilon_N = 2^{-r} \varepsilon_M$, $\log_N := 1 + \log(1/\varepsilon_N)$. The lattice Hamiltonian is $K_{N,L}$ (model sheet (3.7), $g_N \equiv 1$), with eigenvalues $E_0(N) < E_1(N) \leq \dots$, ground state $\omega_N := \omega_{N,L,\lambda,1}$ (model sheet (3.18)); the continuum Hamiltonian is K_L (V3, Theorem 4.1) with eigenvalues $\mu_0 < \mu_1 \leq \dots$, gap $\gamma_L > 0$, and ground state $\omega_{L,\lambda}^{\text{ct}}$ (V3, Theorem 7.1). Write

$$e_0(N) := \frac{1}{2} \sum_{k \in \Gamma_N} \gamma_N(k), \quad \tilde{E}_j(N) := E_j(N) - e_0(N), \quad D_N(t) := \sum_{j \geq 1} e^{-t(E_j(N) - E_0(N))},$$

($e_0(N)$ is the ground energy of the model-sheet free operator $H_{N,L}^{(0)}$, whose normal ordering is the only normalization used below; D_N is shift-invariant), and $D_\infty(t) := \sum_{j \geq 1} e^{-t(\mu_j - \mu_0)} < \infty$ (V3, Theorem 5.2). The thermal functionals are

$$F_N(t; \xi) := \frac{\text{Tr}(e^{-tK_{N,L}} W_N(R_M^N \xi))}{\text{Tr} e^{-tK_{N,L}}}, \quad F_\infty(t; \xi) := \frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr} e^{-tK_L}}, \quad (1)$$

the second well defined by V3, Theorem 5.2; $|F_N(t; \xi)| \leq 1$ for all $N \leq \infty$ ($|\text{Tr}(TW)| \leq \|W\| \text{Tr} T$ for $T \geq 0$ trace class, and the Weyl operators are unitary).

1.2 Deliverables

- (A6) For every fixed $t \geq 1/L$ (equivalently, every $t \in [t_0, T]$ with $t_0 L \geq 1$): $F_N(t; \xi) \rightarrow F_\infty(t; \xi)$, with an explicit rate (Theorem 4.1).
- (A7) $\text{Tr} e^{-t(K_{N,L} - e_0(N))} \rightarrow \text{Tr} e^{-tK_L}$ for every $t \geq t_*$, $t_* > 0$ fixed; $\inf_N \tilde{E}_0(N) > -\infty$; consequently (scoping Lemma 3.2, whose hypothesis (i) is likewise only for $t \geq t_*$) $\tilde{E}_j(N) \rightarrow \mu_j$ for every j , $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$, and $D_N(t) \rightarrow D_\infty(t)$ for every $t \geq t_*$ (Theorem 5.2).
- (A8) The gate (II.11) holds pointwise for every M, ξ ; the Weyl criterion and the identification corollary deliver the full-sequence, identified, fixed-coarse theorem, $\mathbb{Z}_2$ -invariantly (Theorems 6.2 and 6.4).

1.3 Imports

All are program-internal; no new external source is consumed.

- *Scoping note* (`gate_III11_scoping`): Lemma 3.1 (interchange of $t \rightarrow \infty$ and $N \rightarrow \infty$; hypotheses (i)–(iii)), Lemma 3.2 (Laplace transfer of spectra; hypotheses (i)–(ii)), both proved there.
- *V1* (`v1_thermal_representations`): Theorem 5.4 with display (14) (lattice mean-shift representation; the path integrand read as the interaction $\mathcal{U}_N$ — see the erratum note, Remark 1.1), eq. (12) (the action S_N), Remark 5.5 (the ensemble), Lemma 7.1, display (18) (the momentum-weight derivative bound, uniform on $t \geq t_0$; its display (19), the position-weight analogue, is context and is not consumed here).

- **V2 (v2_interaction_comparison)**: Definition 1.1 and Prop. 1.2 (the coupling; laws), Definition 2.2 and Lemma 2.1 ($\mathcal{U}_{N,t}$, $\mathcal{U}_t$, recombination), Theorem 4.3 (A4(a): $\|\mathcal{U}_{N,t} - \mathcal{U}_t\|_{L^2} \leq C(1+t)(1+L)\varepsilon_N \log_N^{3/2}$), Lemma 3.2 with eq. (9) and the k_0 -count eq. (13), eq. (18) (filter tail), Definition 6.2 and Lemma 6.3 (shift fields $h^{(N)}$, $h^{(\infty)}$ and envelopes), Lemma 6.4 (folded/exact-output/high-output symbols, including the representative map), Prop. 6.5 (shifted-interaction comparison with the real asymmetric-band surrogate, exact $A_{\times,j}$ partition, and separate physical-edge bound), Theorem 7.2 (uniform Nelson bounds, plain and shifted).
- **V3 (v3_continuum_package)**: Theorems 4.1, 5.2, 6.2, 7.1 (the A1 package) and Corollary 6.3 (A2(b): the continuum mean-shift representation); Theorem 3.1(ii),(v)–(vi) with Remark 3.2 (the finite-mode package for the abstract oscillator class — applied below *also* to the lattice system, whose membership in the class is the model-sheet §3 proposition); Proposition 2.6(iii) and Lemma 6.1(iv) (the *continuum* uniform Nelson bounds, plain and shifted, used in Lemma 3.3); Remark 4.2 (the A1 normalization of $\omega_{L,\lambda}^{\text{ct}}$, consumed in Theorem 6.2); §1 (the gate endpoint convention $(a,b) = (0,0)$).
- *Free-rate document*: Lemma 2.1 (envelope) and Lemma 3.1 (dispersion) — consumed through V2’s re-packaging and, in Lemma 5.1, directly; the band bound $\gamma_N \geq \gamma_- := \max(m, \frac{2}{\pi}|k|)$ used there is from V2, Lemma 2.1’s proof. (Free-rate Lemma 2.2 is *not* consumed here: the filter tail used below is V2’s own eq. (18), proved there from the QMF telescope.)
- *Weyl-reduction document (lampoart_part_ii_weyl_reduction)*: Theorem 1.1 (full-sequence Weyl criterion), Corollary 3.1 (identification on one common algebra), Proposition 4.1 (compatible symmetry), all proved there.

Remark 1.1 (erratum register for upstream displays). Twelve upstream errata and one normalization amendment are registered here and corrected in place; items (3)–(6) were added after the round-5 adversarial pass and items (7)–(10) after the round-6 pass (13 Aug 2026). Items (1), (1’), (1’’) do not propagate (item (1) is not merely notational: it replaces an import, and the replacement is proved in V3, Theorem 3.1(ii)); item (1’’) is a substantive correction that *did* propagate, and its three propagations (V2 Corollary 5.2 with Remark 5.3, V3 §1’s endpoint, the scoping note’s clause A4(b)) are recorded with it; item (2) is a normalization amendment. (1) *V1, Theorem 5.4 — the reference measure (upgraded 12 Aug 2026, after the adversarial pass)*. The statement originally introduced “ U the full multiplication potential of $K_{N,L}$ ” for the path integrand of display (14), and V1’s import (I1) cited the model sheet’s formula (3.16). Those are two different Feynman–Kac formulas: (3.16) has the *flat* Gaussian kernel, the Brownian bridge, and the full potential, whereas V1’s jump chain (its Lemmas 5.1–5.3) is built from *Mehler* kernels, for which the integrand must be the interaction $\mathcal{V}_N = \lambda \varepsilon_N \sum_x \varphi_x^4$; alone, the harmonic part being carried by the reference. Reading U as $\mathcal{V}_N$ repairs the *statements* (14), (15) and the phase cancellation, but not the *import*: the harmonic-reference formula is not model-sheet (3.16). It is, however, available in the corpus — V3, Theorem 3.1(ii) proves exactly it (Mehler reference, Trotter for bounded truncations, monotone convergence) for the abstract finite-mode class containing the lattice system (V3, Remark 3.2). V1’s import (I1) has accordingly been *restated* as the harmonic-reference formula with that proof. Two caveats belong to the restatement and are recorded in (I1) itself: V3 3.1(ii) states the identity in the Gaussian (μ -) representation, so passing to V1’s Lebesgue-kernel form uses the ground-state (Doob) conjugation $k_t^{(0)} = e^{-tE_N^{(0)}} \Omega_0 \otimes \Omega_0 \cdot M_t$ (the induced bridges coincide); and the acyclicity note is not that V3 3.1(ii) consumes *nothing* from V1 — it does invoke V1’s Mehler kernel, Lemma 5.1 — but that Lemma

5.1 is proved independently of (I1), so the order V1 Lem. 5.1 $\rightarrow$ V3 3.1(ii) $\rightarrow$ V1 §3 and Lemmas 5.2–5.3, Theorem 5.4 $\rightarrow$ V3 3.1(v)–(vi) is acyclic. (1') *V1, display (15)*. As printed, the Wick-binomial shift carried a spurious term $-6c_{N,L}(2H_x\eta_x + H_x^2)$ on top of the Wick-power form: by the Appell property, $:(\eta + H)^4:_c - :\eta^4:_c = 4H:\eta^3:_c + 6H^2:\eta^2:_c + 4H^3\eta + H^4$ exactly, the $-6c[2H\eta + H^2]$ of the bare expansion being consumed by $\eta^3 = :\eta^3:_c + 3c\eta$, $\eta^2 = :\eta^2:_c + c$. Display (15) and its proof have been corrected; every downstream user (V2, Prop. 6.5; V3, Theorem 3.1(vi); §2 here) already used the correct Appell form, so nothing propagated. (1'') *Symbol-difference exponent*. V2, eq. (9) has decay exponent -2 (one envelope factor times one factor $(m^2 + |\kappa|^2)^{-1}$); two places transcribed it as -3 (V2, Lemma 6.4's proof; Lemma 3.2(ii) here). Both have been recomputed with the correct exponent: V2, Lemma 6.4 keeps its bound $B'_j\varepsilon_N^2\log_N^2$ (via the flat bound $|\hat{c}_N - \hat{c}_\infty| \leq C\varepsilon_N^2$ and a band count), and Lemma 3.2 keeps $C_\xi\varepsilon_N^2$ (the $D4$ envelope makes the mode sum converge). (1''') *The convention lock (round 2, breaking)*. V2's Corollary 5.2 asserted that, relative to the truncation convention, the gate's continuum polynomial is $\varphi^4 - 6\delta_\infty\varphi^2 + 3\delta_\infty^2$, and V3 §1 accordingly set the endpoint at $(a, b) = (-6\delta_\infty, 3\delta_\infty^2)$. This is *false*: the lattice and continuum fields' equal-point variances differ by the same δ_∞ as their Wick constants, so the offsets cancel and no counterterm survives. V2, Corollary 5.2 has been restated (with Remark 5.3), V3 §1 now fixes $(a, b) = (0, 0)$, and the scoping note's clause A4(b) — whose own sign has since been corrected to $-6\delta_\infty$, matching the superseded form of V2, Corollary 5.2, and which carried the same underlying trap — is marked superseded, with outcome $(a, b) = (0, 0)$. Nothing in the present document changes: A6–A8 were written for the $(a, b) = (0, 0)$ pair throughout, as the Jensen value in Lemma 3.3(i) shows. (2) *Scoping note, Statement A7*: the parenthetical free-factor display $\prod_k(2\sinh(t\gamma_N(k)/2))^{-1} \rightarrow \prod_k(2\sinh(t\gamma(k)/2))^{-1}$ is vacuous as written: both sides contain the divergent ground-energy factors $e^{-te_0(N)}$, and the infinite product on the right diverges to 0 ($\sum_k \gamma(k) = \infty$). A7 is executed below in the normal-ordered normalization $\tilde{K}_N := K_{N,L} - e_0(N)$, for which the free factor is $\prod_{k \in \Gamma_N}(1 - e^{-t\gamma_N(k)})^{-1} \rightarrow Z_0(t)$ (Lemma 5.1); every quantity consumed by the scoping lemmas (eigenvalue differences, D_N , gaps) is invariant under the c -number shift. Recorded as an amendment in **strategy.md**. (3) *V2, Definition 1.1 — the edge column (round 5, moderate)*. The same-noise coupling defined η_N by restricting the D_∞ -indexed Hermitian family G to D_N . Since Γ_N is asymmetric, $D_N \neq -D_N$: the column $k_e = -\pi/\varepsilon_N$ lies in D_N while $-k_e$ does not, the constraint $G(-\kappa) = \overline{G(\kappa)}$ does not close there, and η_N was consequently *not real* and did *not* carry the law $\mu_{N,t}^{\text{per}}$; $\mathbb{E}[\eta_N(z)\eta_N(z')]$ omitted the entire edge column, a defect of size $\varepsilon_N/(8L)$ at coincident points — one power of ε_N larger than every term tracked in V2, §4. This is load-bearing here: Lemma 3.2 closes with $|e^{ia} - e^{ib}| \leq |a - b|$ for *real* a, b , and V2's Lemma 7.1 is a pointwise identity between real numbers. V2, Definition 1.1 now symmetrizes that one column, $G_N(k_0, k_e) := 2^{-1/2}(G(k_0, k_e) + G(k_0, -k_e))$, which is legitimate because $e^{ik_ex} = e^{-ik_ex}$ on Λ_N and $\gamma_N(k_e) = \gamma_N(-k_e)$; V2, Proposition 1.2(i) then holds verbatim, (ii) acquires an explicit edge remainder R_N , and (iii) records that every second- and fourth-moment sum of V2, §4 changes by $O(t\varepsilon_N^3)$. No statement and no rate in V2 or here changes. (4) *Scoping note, Statement A7 — unshifted symbols (round 5)*. The normal-ordering amendment of item (2) had been inserted, but A7's *conclusions* were left in the un-normal-ordered symbols, in which two of them are false: $e_0(N) \geq r_N m \rightarrow \infty$ forces $Z_N(t) \rightarrow 0 \neq Z_\infty(t)$ and $E_j(N) \rightarrow +\infty \neq \mu_j$. They are now stated in the tilded symbols that Theorem 5.2 actually proves; only the shift-invariant conclusions ($E_1 - E_0 \rightarrow \gamma_L$, $D_N \rightarrow D_\infty$) were ever consumed by A8. (5) *Scoping note, Lemma 3.2 — the small- t hypothesis and the density step (round 5)*. Hypothesis (i) demanded $Z_N(t) \rightarrow Z_\infty(t)$ for *every* $t > 0$, which Theorem 5.2(a) would have had to supply by invoking V2, Theorem 4.3 at arbitrarily small t — where that theorem's standing hypothesis $tL \geq 1$ fails and its constant genuinely blows up (Σ_1 saturates as $t \downarrow 0$ while $tL \rightarrow 0$). Hypothesis (i) is now required

only for $t \geq t_*$ with $t_* > 0$ fixed, which still yields Stone–Weierstrass density (the span of $\{e^{-tE} : t \geq t_*\}$ is an algebra), and Theorem 5.2 is stated for $t \geq t_*$; t_1 is the only value A8 consumes. Independently, that lemma’s proof approximated against ν_N itself, which has infinite total mass, so $\|f - g\|_\infty < \varepsilon$ controlled nothing; the weighted argument against $e^{-t_*E} d\nu_N$, previously a gloss, is now the proof. (6) *Scoping note, Statement A3(b) — a false tail bound (round 5)*. The justification clause used $\coth(t\gamma/2) - 1 \leq 2e^{-t_0\gamma}$, which is false for all $t_0, \gamma > 0$ (since $2/(e^x - 1) - 2e^{-x} = 2/[e^x(e^x - 1)] > 0$) and fails by a factor $\sim 1/(t_0\gamma)$. Corrected to $\coth(t\gamma/2) - 1 \leq 2e^{-t_0\gamma}/(1 - e^{-t_0m})$. The executed proof (V1, Theorem 7.2) never used the false form. (7) *V2, §§4 and 6 — the amendment was not propagated (round 6)*. Round 5 amended V2’s Definition 1.1 but left §4 and §6 standing against the old coupling. Three consequences, all now corrected in V2: (a) the mixed pairing does *not* run over D_N with symbol $\hat{c}_\times$ but over the *symmetric* band $\tilde{D}_N$ with the edge weight $\chi(\pm\pi/\varepsilon_N) = 2^{-1/2}$ — it is false that modes of η outside D_N have zero covariance with η_N , since the symmetrization is exactly what couples $+\pi/\varepsilon_N$ to η_N ; (b) the blanket claim that this changes every second- and fourth-moment sum by $O(t\varepsilon_N^3)$ is wrong on both counts — the second-moment sums are unchanged *exactly*, and the fourth-moment cross sum computed on D_N^4 rather than on $(\tilde{D}_N, \chi)$ has one-, two-, and four-edge sectors. The former two are $O(t\varepsilon_N^3 \log_N^2)$; the formerly omitted four-edge sector is

$$4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_i \hat{c}_\times(k_{0,i}, \pi/\varepsilon_N) \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5.$$

Exactly three edge factors are impossible, but four are not. V2 now tracks the total correction as $\mathcal{E}_N^{\text{edge}}$ in V2’s displays `eq:Eedge=eq:Eedge-bound` and carries it into the fourth-chaos norm; (c) V2’s §6 has now been re-derived with three different multipliers: the folded lattice symbol $K_{N,j}^{\text{fold}}$, the exact-output mixed symbol $K_{\times,j}$, and the continuum symbol $K_{\infty,j}$. Since the asymmetric-band mixed kernel can be complex at the edge, the scalar surrogate is defined as the real part of that pairing. Its shifted quadratic difference is the exact four-term partition containing the displayed mixed-alias term $A_{\times,j}$; the former unnamed remainder has been deleted. The physical edge kernel is bounded separately before the Fourier partition. These repairs preserve Proposition 6.5’s rate, so Lemma 3.3(iii) and A6 remain valid with an actual proved input rather than the superseded decomposition. (8) *The D4 filter annihilates the edge column (round 6, new; scope corrected in round 7)*. For the D4 mask $m_0(\pi) = 0$, whence $P_r(\pm\pi 2^r) = 0$ and the coarse coefficients a_N, b_N *vanish* on the edge column for every $r \geq 1$. This was noticed in no earlier round. It is what makes $h^{(N)}$ real and what licenses Lemma 3.2 here (for $r \geq 1$; the case $r = 0$ is disposed of separately there, the identity being genuinely false at $r = 0$). It does *not* remove the edge remainder from V2’s §6 at all — neither in general, as round 6 first claimed, nor for $j = 3$, as round 7 then claimed. Round 8 located the actual mechanism: $R_N = \sigma\Psi$ with $\sigma(x) = (-1)^{x/\varepsilon_N}$ on Λ_N , so $\sigma^2 \equiv 1$ and the expansion of $(C_\times + R_N)^j - C_\times^j$ contains σ -free terms as soon as $j \geq 2$; those are supported on the whole j -fold sumset and no vanishing of $a_{k_e}^{(N)}$ touches them. Only a kernel entering *linearly* is edge-supported — which is Lemma 3.2 here, and not any chaos of V2, Proposition 6.5, whose remainder is bounded uniformly in $j \in \{1, 2, 3\}$. Recorded in V2 as the remark following Definition 6.2, clause (b). Three successive rounds asserted a vanishing here; the lesson recorded in `strategy.md` is that a convolution or a power of vanishing quantities does not vanish. (9) *The standing hypothesis $tL \geq 1$ (round 6)*. Round 5 attached it to V2’s Theorem 4.3 only. It is consumed just as much by V2’s Lemma 4.2, Lemma 6.4, Proposition 6.5 and Theorem 7.2, and here by Lemma 3.3 and Theorem 4.1; erratum item (5) was wrong to say Theorem 5.2(a) is the only consumer. V2 now carries it as the standing convention $(\star)$ of its §1, and Lemma 3.3, Theorem 4.1 and Theorem 6.2 state it explicitly.

Correspondingly Theorem 6.2 no longer fixes $t_0 = t_1 = 1$ — inadmissible when $L < 1$, since $t_* L \geq 1$ forces $t_* \geq 1/L$ — but $t_0 = t_1 = t_* := \max(1, 1/L)$. Round 7 added two further consumers that had been missed: Theorem 5.2(a) invoked Lemma 3.3 at $t_0 := t/2$, which violates $(\star)$ on the whole window $t \in [t_*, 2/L)$ — exactly where the gate consumes it — and is now $t_0 := \max(t/2, 1/L)$; and V3, which imports seven V2 results each stated only under $(\star)$ but nowhere assumed it, now carries it as a standing convention of its §1. (10) *Scoping note, Lemma 3.2 — conclusion still quantified over all t (round 6, breaking as stated).* After the round-5 weakening of hypothesis (i) to $t \geq t_*$, the conclusion $D_N(t) \rightarrow D_\infty(t)$ was left quantified over every $t > 0$, where it is false: with $\nu_N = \sum_{j \geq 0} \delta_j + \lceil e^N/N \rceil \delta_N$ all hypotheses hold while $D_N(\frac{1}{2}) - D_\infty(\frac{1}{2}) \sim e^{N/2}/N \rightarrow \infty$. Restricted to $t \geq t_*$, with the counterexample recorded there. Only $t_1 \geq t_*$ is consumed, by Theorem 5.2(c).

2 The two representations on the coupling space

Work on the V2 coupling space (V2, Definition 1.1): one Gaussian family realizes the lattice field η_N (law = V1's periodic ensemble $\mu_{N,t}^{\text{per}}$; V2, Prop. 1.2) and the continuum field η (law = $\mu_{t,L}^{\text{per}}$) simultaneously. Let $\mathcal{U}_{N,t}, \mathcal{U}_t$ be the V2 interactions (V2, Definition 2.2), $h^{(N)}, h^{(\infty)}$ the shift fields of the symbol ξ (V2, Definition 6.2; $h^{(N)}$ is V1's interpolant $\mathbf{H}_p$ in the original coordinates, $h^{(\infty)}$ its continuum counterpart), and define the amplitude collapses

$$b_N(k) := \varepsilon_M^{1/2} P_r(\varepsilon_M k) \widehat{p}_M(k), \quad a_N(k) := \varepsilon_M^{1/2} P_r(\varepsilon_M k) \widehat{q}_M(k) \quad (k \in \Gamma_N), \quad (2)$$

and b_∞, a_∞ with P_∞ on Γ_∞ (the $2^{r/2}$ -normalizations cancel as in V2, Definition 6.2: $\varepsilon_N^{1/2} \widehat{p}_N(k) = b_N(k)$, etc.). The actions and source phases are

$$S_N = \frac{1}{4 \cdot 2L} \sum_{k \in \Gamma_N} \gamma_N(k) \coth\left(\frac{t\gamma_N(k)}{2}\right) |b_N(k)|^2, \quad \Phi_N(J_q) = \varepsilon_N \sum_{x \in \Lambda_N} q_N(x) \eta_N(0, x), \quad (3)$$

and $S_\infty, \Phi(J_q) = \int_{\mathbb{T}_L} q_\infty(x) \eta(0, x) dx$ analogously (S_N is V1, eq. (12), rewritten with $(2r_N)^{-1} |\widehat{p}_N|^2 = (2L)^{-1} |b_N|^2$; S_∞ is V3, §6). By V1, Theorem 5.4 (with Remark 1.1(1)) and V3, Corollary 6.3:

$$F_N(t; \xi) = e^{-S_N} \frac{\mathcal{N}_N(t; \xi)}{\mathcal{E}_N(t)}, \quad F_\infty(t; \xi) = e^{-S_\infty} \frac{\mathcal{N}_\infty(t; \xi)}{\mathcal{E}_\infty(t)}, \quad (4)$$

where, as expectations over the coupling space,

$$\begin{aligned} \mathcal{N}_N &:= \mathbb{E}[e^{i\Phi_N(J_q)} e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})}], & \mathcal{E}_N &:= \mathbb{E}[e^{-\mathcal{U}_{N,t}}], \\ \mathcal{N}_\infty &:= \mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta + h^{(\infty)})}], & \mathcal{E}_\infty &:= \mathbb{E}[e^{-\mathcal{U}_t}]. \end{aligned}$$

Throughout, C_ξ denotes a finite constant depending on $(m, L, \lambda, t_0, T, M, \|\widehat{q}_M\|_\infty, \|\widehat{p}_M\|_\infty)$ but not on N or $t \in [t_0, T]$; $\eta = 2 - \log_2 3$ and $\bar{w} := \coth(t_0 m/2)$.

3 Two comparison lemmas

Lemma 3.1 (action prefactors). *For $t \in [t_0, T]$ and $N = M + r$,*

$$0 \leq S_N, \quad S_\infty \leq C_\xi, \quad |S_N - S_\infty| \leq C_\xi 2^{-r\eta/2}, \quad |e^{-S_N} - e^{-S_\infty}| \leq |S_N - S_\infty|.$$

Proof. Write $w_\bullet^p(k) := \gamma_\bullet(k) \coth(t\gamma_\bullet(k)/2)$, so $4 \cdot 2L S_\bullet = \sum w_\bullet^p |b_\bullet|^2$. Boundedness: $w^p \leq \bar{w} \gamma(k)$, $|b_\bullet(k)|^2 \leq C\varepsilon_M \|\hat{p}_M\|_\infty^2 (1 + \varepsilon_M |k|)^{-2-\eta}$ (D4 envelope, V2 §6), and $\sum_k \gamma(k)(1 + \varepsilon_M |k|)^{-2-\eta} < \infty$ since the summand decays like $|k|^{-1-\eta}$. For the difference, split at $\ell_r := 2^{r/2}$ in the variable $\ell := \varepsilon_M |k|$, and use, with $s_* := \sup_{x \geq t_0 m/2} x / \sinh^2 x$:

Low modes ($\ell \leq \ell_r$). Telescope $|w_N^p |b_N|^2 - w^p |b_\infty|^2| \leq w_N^p ||b_N|^2 - |b_\infty|^2| + |w_N^p - w^p| |b_\infty|^2$. By V1, eq. (18), $|w_N^p - w^p| \leq (\bar{w} + s_*) \varepsilon_N^2 k^4 / (24m)$; summing against the envelope, $\sum_{\ell \leq \ell_r} \varepsilon_N^2 k^4 (1 + \ell)^{-2-\eta} \leq C(M) 4^{-r} \ell_r^{3-\eta} = C 2^{-r(1+\eta)/2} (\int_0^X \ell^{2-\eta} d\ell = X^{3-\eta}/(3-\eta))$, with $\varepsilon_N^2 k^4 = 4^{-r} \ell^4 \varepsilon_M^{-2}$; the mode density in the variable $\ell = \varepsilon_M |k|$ is the constant $2L/(\pi \varepsilon_M)$ and supplies no factor of ℓ — the integrand $\ell^{2-\eta}$ is already the large- ℓ behaviour of the summand $\ell^4 (1 + \ell)^{-2-\eta}$. By the filter-tail bound (V2, eq. (18)): $|b_N - b_\infty|(k) \leq \varepsilon_M^{1/2} \|\hat{p}_M\|_\infty C_D^{1/2} L_{m_0} 2^{-r} \ell (1 + \ell)^{-1-\eta/2}$, so $||b_N|^2 - |b_\infty|^2| \leq |b_N - b_\infty|(|b_N| + |b_\infty|) \leq C\varepsilon_M \|\hat{p}_M\|_\infty^2 2^{-r} \ell (1 + \ell)^{-2-\eta}$, and with $w_N^p \leq \bar{w}(m + |k|) \leq \bar{w}(m + \ell/\varepsilon_M)$: $\sum_{\ell \leq \ell_r} \leq C(M) 2^{-r} \int_0^{\ell_r} \ell^2 (1 + \ell)^{-2-\eta} d\ell \leq C 2^{-r} \ell_r^{1-\eta} = C 2^{-r(1+\eta)/2}$.

High modes ($\ell > \ell_r$, including $|k| > \pi/\varepsilon_N$). Bound both terms separately by the envelope: $\sum_{\ell > \ell_r} \bar{w} \gamma(k) C\varepsilon_M (1 + \ell)^{-2-\eta} \leq C(M) \int_{\ell_r}^\infty \ell (1 + \ell)^{-2-\eta} d\ell \leq C \ell_r^{-\eta} = C 2^{-r\eta/2}$. Collecting, $|S_N - S_\infty| \leq C_\xi (2^{-r(1+\eta)/2} + 2^{-r\eta/2}) \leq C_\xi 2^{-r\eta/2}$. The last display of the lemma is the mean value theorem on $[0, \infty)$. $\square$

Lemma 3.2 (marked-slice source phases). *On the coupling space, $\Phi_N(J_q) - \Phi(J_q)$ is a centered Gaussian variable with*

$$\mathbb{E}[(\Phi_N(J_q) - \Phi(J_q))^2] \leq C_\xi 2^{-r\eta/2}, \quad \|e^{i\Phi_N(J_q)} - e^{i\Phi(J_q)}\|_{L^2(\mathbb{P})} \leq C_\xi 2^{-r\eta/4}.$$

Proof. By V2, Definition 1.1 (in its amended form, with the symmetrized family G_N),

$$\begin{aligned} \eta_N(0, x) &= (2Lt)^{-1/2} \sum_{\kappa \in D_N} G_N(\kappa) \hat{c}_N(\kappa)^{1/2} e^{ikx}, \\ \varepsilon_N \sum_x q_N(x) e^{ikx} &= \varepsilon_N^{1/2} \hat{q}_N(-k) = a_N(-k), \end{aligned}$$

so $\Phi_N(J_q)$ is a real Gaussian. Assume $r \geq 1$ for the rest of the proof; the case $r = 0$ is disposed of at the end. Then the edge column drops out entirely: $a_N(\pm\pi/\varepsilon_N) = 0$, because $a_N \propto P_r(\varepsilon_M k)$ and $P_r(\pm\pi 2^r) = 0$, the mask factor $m_0(\pm\pi)$ vanishing (V2, the remark following Definition 6.2). Hence G_N may be replaced by G in the expansion below without changing anything, the two differing only on that column, and

$$\begin{aligned} \Phi_N(J_q) &= (2Lt)^{-1/2} \sum_{\kappa \in D_N} G(\kappa) \hat{c}_N(\kappa)^{1/2} a_N(-k), \\ \Phi(J_q) &= (2Lt)^{-1/2} \sum_{\kappa \in D_\infty} G(\kappa) \hat{c}_\infty(\kappa)^{1/2} a_\infty(-k), \end{aligned}$$

whence, by orthonormality of the $G(\kappa)$,

$$\mathbb{E}[(\Phi_N - \Phi)^2] = \frac{1}{2Lt} \left[\sum_{\kappa \in D_N} |\hat{c}_N^{1/2} a_N - \hat{c}_\infty^{1/2} a_\infty|^2(-k) + \sum_{\kappa \in D_\infty \setminus D_N} \hat{c}_\infty |a_\infty|^2 \right].$$

Split $\hat{c}_N^{1/2} a_N - \hat{c}_\infty^{1/2} a_\infty = \hat{c}_N^{1/2} (a_N - a_\infty) + (\hat{c}_N^{1/2} - \hat{c}_\infty^{1/2}) a_\infty$ and use $(x + y)^2 \leq 2x^2 + 2y^2$: (i) $t^{-1} \sum_{k_0} \hat{c}_N = \coth(t\gamma_N/2)/(2\gamma_N) \leq \bar{w}/2m$ (V2, eq. (2) at $\tau = 0$), and $|a_N - a_\infty|^2 \leq C\varepsilon_M \|\hat{q}_M\|_\infty^2 4^{-r} \ell^2 (1 + \ell)^{-2-\eta}$ (filter tail, as in Lemma 3.1); the mode sum gives $C(M) 4^{-r} \int_0^{\pi 2^r} \ell^2 (1 + \ell)^{-2-\eta} d\ell \leq C 2^{-r(1+\eta)}$. (ii) $(\hat{c}_N^{1/2} - \hat{c}_\infty^{1/2})^2 \leq |\hat{c}_N - \hat{c}_\infty|$, and V2, eq. (9)

reads $|\hat{c}_N - \hat{c}_\infty|(\kappa) \leq \frac{\pi^6}{768} \varepsilon_N^2 k^4 (m^2 + |\kappa|^2)^{-2}$ (decay exponent -2 : one factor $\hat{c}_{\text{env}}$, one factor $(m^2 + |\kappa|^2)^{-1}$), so the k_0 -count V2, eq. (13) applies at $s = 2$ and gives

$$t^{-1} \sum_{k_0} |\hat{c}_N - \hat{c}_\infty| \leq C(m, t_0) \varepsilon_N^2 \frac{k^4}{(m^2 + k^2)^{3/2}} \leq C(m, t_0) \varepsilon_N^2 (m + |k|);$$

against the envelope $|a_\infty(k)|^2 \leq C \varepsilon_M \|\hat{q}_M\|_\infty^2 (1 + \ell)^{-2-\eta}$, $\ell = \varepsilon_M |k|$, the summand is $O(\ell^{-1-\eta})$, so the mode sum converges and the term is $\leq C_\xi \varepsilon_N^2$. (iii) The missing modes: $t^{-1} \sum_{k_0} \hat{c}_\infty = \coth(t\gamma/2)/(2\gamma) \leq \bar{w}/2m$ and the envelope tail over $\Gamma_\infty \setminus \Gamma_N$, i.e. $\ell \geq \pi 2^r$, is $\leq C 2^{-r(1+\eta)}$ (the extra edge mode $k = +\pi/\varepsilon_N$ obeys the same envelope, so no constant changes). Collecting: $\mathbb{E}[(\Phi_N - \Phi)^2] \leq C_\xi (2^{-r(1+\eta)} + \varepsilon_N^2) \leq C_\xi 2^{-r\eta/2}$ (a deliberately crude common exponent). The second display follows from $|e^{ia} - e^{ib}| \leq |a - b|$ and the L^2 bound.

Finally $r = 0$, where $P_0 \equiv 1$ and $a_N(\pm\pi/\varepsilon_N)$ need not vanish, so the replacement of G_N by G is not licensed and the displayed variance identity is genuinely false (the two sides differ by $(2 - 2\sqrt{2})\varepsilon_M |\hat{q}_M(\pi/\varepsilon_M)|^2 \text{Re } P_\infty(-\pi) (2Lt)^{-1} \sum_{k_0} \hat{c}_\times(k_0, k_e) \neq 0$, since $\text{Re } P_\infty(-\pi) = -0.4919\dots$). Nothing is lost: at $r = 0$ the two fields are indexed by the single fixed scale M , and $\mathbb{E}[(\Phi_N - \Phi)^2] \leq 2\mathbb{E}[\Phi_M^2] + 2\mathbb{E}[\Phi^2] \leq C_\xi$, both terms being finite by the envelope of V2, Lemma 3.1(i); enlarging C_ξ by that one fixed amount gives the stated bound at $r = 0$ as well. $\square$

Lemma 3.3 (numerators, denominators, and their limits). *For $t \in [t_0, T]$ with $t_0 L \geq 1$ (V2's standing hypothesis $(\star)$, inherited through V2, Theorem 4.3 and Proposition 6.5, which clauses (ii),(iii) consume), and with $\kappa_1 := \min\{\eta/4, (1+\eta)/4\} = \eta/4$: (i) $\mathcal{E}_N(t) \geq e^{-C_0} > 0$ and $\mathcal{E}_\infty(t) \geq e^{-C_0} > 0$ with $\sup_N \mathbb{E}[\mathcal{U}_{N,t}] \vee \mathbb{E}[\mathcal{U}_t] \leq 3\lambda \bar{d}^2 2LT =: C_0 < \infty$; (ii) $|\mathcal{E}_N(t) - \mathcal{E}_\infty(t)| \leq C(1+t)(1+L) \varepsilon_N \log_N^{3/2}$; (iii) $|\mathcal{N}_N(t; \xi) - \mathcal{N}_\infty(t; \xi)| \leq C_\xi (2^{-r\kappa_1} + \varepsilon_N \log_N^{3/2})$, and $|\mathcal{N}_\bullet| \leq \sup_{N \leq \infty} \|e^{-\mathcal{U}_{\bullet, t}(\cdot + h(\bullet))}\|_{L^1} =: B_1 < \infty$, the supremum running over the lattice family and its continuum member.*

Proof. Throughout, the *lattice* uniform Nelson bounds are V2, Theorem 7.2, and their *continuum* counterparts $\|e^{-\mathcal{U}_t}\|_{L^p}$, $\|e^{-\mathcal{U}_t(\cdot + h(\infty))}\|_{L^p}$ are V3, Prop. 2.6(iii) and Lemma 6.1(iv); we cite the pair as “the uniform Nelson bounds”. (i) Jensen: $\mathcal{E} \geq e^{-\mathbb{E}[\mathcal{U}]}$, and by V2, Definition 2.2 the ensemble-Wick chaoses are centered, so $\mathbb{E}[\mathcal{U}_{N,t}] = 3\lambda d_N(t)^2 \cdot 2Lt \leq 3\lambda \bar{d}^2 2Lt$ (V2, Lemma 2.1); likewise in the limit. (ii) $|e^{-\alpha} - e^{-\beta}| \leq |\alpha - \beta|(e^{-\alpha} + e^{-\beta})$, Hölder, V2, Theorem 4.3 and the uniform Nelson bounds: $|\mathcal{E}_N - \mathcal{E}_\infty| \leq \|\mathcal{U}_{N,t} - \mathcal{U}_t\|_{L^2} \|e^{-\mathcal{U}_{N,t}} + e^{-\mathcal{U}_t}\|_{L^2}$. (iii) Split

$$|\mathcal{N}_N - \mathcal{N}_\infty| \leq \mathbb{E} \left[\left| e^{i\Phi_N(J_q)} - e^{i\Phi(J_q)} \right| e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})} \right] + \mathbb{E} \left[\left| e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})} - e^{-\mathcal{U}_t(\eta + h^{(\infty)})} \right| \right].$$

The first term is $\leq \|e^{i\Phi_N} - e^{i\Phi}\|_{L^2} \|e^{-\mathcal{U}_{N,t}(\cdot + h^{(N)})}\|_{L^2} \leq C_\xi 2^{-r\eta/4} B_2$ (Lemma 3.2; B_2 from V2, Theorem 7.2, shifted clause). For the second, decompose the exponent difference as

$$\mathcal{U}_{N,t}(\eta_N + h^{(N)}) - \mathcal{U}_t(\eta + h^{(\infty)}) = [\mathcal{U}_{N,t}(\eta_N) - \mathcal{U}_t(\eta)] + [\Delta\mathcal{U}_N - \Delta\mathcal{U}],$$

with $\Delta\mathcal{U}_N := \mathcal{U}_{N,t}(\cdot + h^{(N)}) - \mathcal{U}_{N,t}(\cdot)$ as in V2, Prop. 6.5; by V2, Theorem 4.3 (contributing $\varepsilon_N \log_N^{3/2}$) and Prop. 6.5 (contributing $2^{-r(1+\eta)/4} + \varepsilon_N \log_N$) its L^2 norm is $\leq C_\xi (2^{-r(1+\eta)/4} + \varepsilon_N \log_N^{3/2})$, and the $|e^{-\alpha} - e^{-\beta}|$ inequality with the shifted uniform Nelson bounds — the *pair* named in the preamble, since β is the continuum member — closes as in (ii). $B_1 < \infty$ holds by the same pair (V2, Theorem 7.2 for $N < \infty$; V3, Lemma 6.1(iv) at $p = 1$ for the continuum member). $\square$

4 A6: fixed- t convergence of the thermal functionals

Theorem 4.1 (A6). *For every $0 < t_0 \leq t \leq T$ with $t_0 L \geq 1$, every M , and every $\xi \in h_{M,L}$,*

$$|F_N(t; \xi) - F_\infty(t; \xi)| \leq C_\xi \left(2^{-r\eta/4} + \varepsilon_N \log_N^{3/2} \right) \xrightarrow{N \rightarrow \infty} 0.$$

In particular $F_N(t; \xi) \rightarrow F_\infty(t; \xi)$ for every fixed $t \geq 1/L$ (choose $t_0 := \max(t/2, 1/L) \leq t \leq T := 2t$). The restriction is real, not formal: the proof consumes V2, Theorem 4.3 and Proposition 6.5 through Lemma 3.3(ii),(iii), and V2's remark following Theorem 4.3 shows the constant there genuinely diverges as $tL \downarrow 0$. Only $t \geq t_ := \max(1, 1/L)$ is consumed downstream (Theorem 6.2).*

Proof. By (4),

$$F_N - F_\infty = (e^{-S_N} - e^{-S_\infty}) \frac{\mathcal{N}_N}{\mathcal{E}_N} + e^{-S_\infty} \frac{\mathcal{N}_N - \mathcal{N}_\infty}{\mathcal{E}_N} + e^{-S_\infty} \mathcal{N}_\infty \frac{\mathcal{E}_\infty - \mathcal{E}_N}{\mathcal{E}_N \mathcal{E}_\infty}.$$

Insert Lemmas 3.1–3.3: each denominator is $\geq e^{-C_0}$, each numerator factor is bounded by B_1 or 1, and the three differences are $O(2^{-r\eta/2})$, $O(2^{-r\eta/4} + \varepsilon_N \log_N^{3/2})$ and $O(\varepsilon_N \log_N^{3/2})$ respectively. $\square$

Remark 4.2. This discharges the scoping note's “conclude by Vitali” step in quantitative form: the elementary inequality $|e^{-\alpha} - e^{-\beta}| \leq |\alpha - \beta|(e^{-\alpha} + e^{-\beta})$ together with the uniform L^p bounds replaces the Vitali convergence theorem, and yields rates. No complex-analytic input is used.

5 A7: partition functions, Laplace transfer, the derived gap

Lemma 5.1 (free products). *For every $t > 0$,*

$$\tilde{Z}_N^{(0)}(t) := \text{Tr } e^{-t(H_{N,L}^{(0)} - e_0(N))} = \prod_{k \in \Gamma_N} (1 - e^{-t\gamma_N(k)})^{-1} \longrightarrow Z_0(t) = \prod_{k \in \Gamma_\infty} (1 - e^{-t\gamma(k)})^{-1},$$

with $|\log \tilde{Z}_N^{(0)}(t) - \log Z_0(t)| \leq C(m, L, t) \varepsilon_N^2$.

Proof. – log of the products differ by

$$\left| \sum_{k \in \Gamma_N} [\log(1 - e^{-t\gamma_N(k)}) - \log(1 - e^{-t\gamma(k)})] \right| + \left| \sum_{k \in \Gamma_\infty \setminus \Gamma_N} \log(1 - e^{-t\gamma(k)}) \right|.$$

For the first sum, the mean value theorem in the variable $y = e^{-t\gamma}$ gives per- k bounds

$$\leq \frac{t e^{-t\gamma_N(k)}}{1 - e^{-tm}} |\gamma(k) - \gamma_N(k)| \leq C(m, t) e^{-t \max(m, (2/\pi)|k|)} \varepsilon_N^2 k^4$$

(free-rate Lemma 3.1 and $\gamma_N \geq \max(m, \frac{2}{\pi}|k|)$), summable to $C\varepsilon_N^2$; the second sum, over $\Gamma_\infty \setminus \Gamma_N = \{k \geq \pi/\varepsilon_N\} \cup \{k < -\pi/\varepsilon_N\}$ — which contains $k = +\pi/\varepsilon_N$, the band being asymmetric — is $\leq \sum_{|k| \geq \pi/\varepsilon_N} \frac{e^{-t\gamma(k)}}{1 - e^{-tm}} \leq C(m, L, t) e^{-2t/\varepsilon_N}$, since $\gamma(k) \geq |k| \geq \pi/\varepsilon_N > 2/\varepsilon_N$ there. $\square$

Theorem 5.2 (A7). *Fix $t_* > 0$ (arbitrary; only $t_* = t_1$ is consumed downstream) and assume $t_* L \geq 1$. (a) For every $t \geq t_*$,*

$$\tilde{Z}_N(t) := \text{Tr } e^{-t(K_{N,L} - e_0(N))} = \tilde{Z}_N^{(0)}(t) \mathcal{E}_N(t) \longrightarrow Z_\infty(t) = Z_0(t) \mathcal{E}_\infty(t) = \text{Tr } e^{-tK_L}.$$

(b) $\inf_N \tilde{E}_0(N) \geq -c_\lambda > -\infty$. (c) *Consequently, by the scoping note's Lemma 3.2 applied to the shifted eigenvalue measures $\tilde{\nu}_N = \sum_j \delta_{\tilde{E}_j(N)}$ and $\nu_\infty = \sum_j \delta_{\mu_j}$: $\tilde{E}_j(N) \rightarrow \mu_j$ for every j ; $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$; and $D_N(t) \rightarrow D_\infty(t)$ for every $t \geq t_*$, in particular $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ at any fixed $t_1 \geq t_*$.*

Proof. (a) The identity $\tilde{Z}_N = \tilde{Z}_N^{(0)} \mathcal{E}_N$ is the sourceless periodic trace identity for the lattice system, i.e. the *finite-mode* instance of V3, Theorem 3.1(v) — the lattice system is purely finite-dimensional, so the identity is the finite-mode statement of that theorem's proof before tensoring with any free tail, with free factor $\tilde{Z}_N^{(0)}$ — applied with the lattice dictionary (dispersions $\gamma_N(k)$, $k \in \Gamma_N$; interaction V_N ; class membership by the model-sheet §3 proposition, cf. V3, Remark 3.2). Its ensemble is $\mu_{N,t}^{\text{per}}$ by V1, Remark 5.5 = V2, Prop. 1.2 — with the edge column symmetrized as in V2, Definition 1.1 and the unnumbered remark following it, without which η_N would be neither real nor of that law — and its path integrand, in ensemble-Wick form, is $\mathcal{U}_{N,t}$ by V2's recombination (Lemma 2.1). Convergence: Lemma 5.1 and Lemma 3.3(ii) (at $t_0 := \max(t/2, 1/L) \leq t$, $T := 2t$ — *not* at $t_0 := t/2$, which would give $t_0 L = tL/2 < 1$ and violate that lemma's own hypothesis $(\star)$ on the whole window $t \in [t_*, 2/L)$, which is where the gate consumes it; the present choice is admissible since $t \geq t_* \geq 1/L$, and is the one Theorem 4.1 already makes. This is where $t \geq t_*$ is needed, V2's comparison carrying the standing hypothesis $tL \geq 1$ — see V2, the remark following Theorem 4.3 — and the scoping note's Lemma 3.2 requiring convergence only for $t \geq t_*$). The right-hand equality is V3, Theorem 6.2(a). (b) $e^{-t_1 \tilde{E}_0(N)} \leq \tilde{Z}_N(t_1) \leq \tilde{Z}_N^{(0)}(t_1) \sup_N \mathcal{E}_N(t_1) < \infty$ uniformly (V2, Theorem 7.2 at $p = 1$ and Lemma 5.1); take logarithms. (c) Hypothesis (i) of the scoping Lemma 3.2 is (a); hypothesis (ii) is (b). The limit spectrum is that of K_L (V3, Theorem 5.2), with μ_0 simple and $\gamma_L = \mu_1 - \mu_0 > 0$ (V3, Theorem 7.1), so the lemma's conclusions read as stated ($E_1(N) - E_0(N) = \tilde{E}_1(N) - \tilde{E}_0(N)$ and D_N are shift-invariant); $D_\infty(t_1) < \infty$ by V3, Theorem 5.2. $\square$

6 A8: assembly — the gate closes

Lemma 6.1 (spectral tail bound). *For every $N \leq \infty$, $t > 0$, and unitary W on the respective space,*

$$\left| \frac{\text{Tr}(e^{-tK}W)}{\text{Tr } e^{-tK}} - \langle \Omega, W\Omega \rangle \right| \leq \frac{2D(t)}{1 + D(t)} \leq 2D(t),$$

where $K \in \{K_{N,L}, K_L\}$, Ω is the (simple) ground vector and $D(t)$ the corresponding tail sum.

Proof. With the orthonormal eigenbasis (ϕ_j) , $\phi_0 = \Omega$, and $Z = \text{Tr } e^{-tK} = e^{-tE_0}(1 + D(t))$:

$$\frac{\text{Tr}(e^{-tK}W)}{Z} - \langle \Omega, W\Omega \rangle = \frac{1}{Z} \sum_{j \geq 1} e^{-tE_j} (\langle \phi_j, W\phi_j \rangle - \langle \Omega, W\Omega \rangle),$$

and $|\langle \phi_j, W\phi_j \rangle - \langle \Omega, W\Omega \rangle| \leq 2$, so the modulus is $\leq 2e^{-tE_0}D(t)/Z = 2D(t)/(1 + D(t))$. For $N = \infty$ this uses V3, Theorems 5.2 and 7.1 (trace class; simple ground state). $\square$

Theorem 6.2 (the gate (II.11)). *For every fixed M and $\xi \in h_{M,L}$,*

$$\lim_{N \rightarrow \infty} \omega_{N,L,\lambda,1}(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))),$$

i.e. display (3.4) of the Weyl-reduction document holds, with $\omega_{P(\phi)_2,L} := \omega_{L,\lambda}^{\text{ct}}$ the A1-normalized continuum state (V3, Remark 4.2).

Proof. Apply the scoping note's Lemma 3.1 with $t_0 := t_* := \max(1, 1/L)$ — so that $t_*L \geq 1$, as Theorem 5.2 requires, and $t_* \geq 1/L$, as Theorem 4.1 requires; the lemma admits any $t_0 > 0$ and any fixed $t_1 \geq t_0$ —

$$a_N(t) := F_N(t; \xi), \quad \ell_N := \omega_N(W_N(R_M^N \xi)), \quad a_\infty(t) := F_\infty(t; \xi), \quad \ell_\infty := \omega_{L,\lambda}^{\text{ct}}(W_{\text{ct}}(R_M^\infty \xi)),$$

$c_0 := 2$. The standing bounds: $|a_N(t)| \leq 1 \leq c_0$; the tail hypothesis — eq. (4) of the scoping note — with constant c_0 is Lemma 6.1 (both for $N < \infty$ and $N = \infty$; $\alpha_M^N(W_M(\xi)) = W_N(R_M^N \xi)$ and $\beta_M(W_M(\xi)) = W_{\text{ct}}(R_M^\infty \xi)$ by the model-sheet D4 embeddings). The three hypotheses: (i) $a_N(t) \rightarrow a_\infty(t)$ for every fixed $t \geq t_0 = t_* \geq 1/L$ is Theorem 4.1; (ii) $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ at $t_1 := t_0 = t_*$ is Theorem 5.2(c), applied with that t_* (admissible, since $t_*L \geq 1$); (iii) $\liminf_N (E_1(N) - E_0(N)) = \gamma_L > 0$ is Theorem 5.2(c). The lemma yields $\ell_N \rightarrow \ell_\infty$, which is the display. $\square$

Lemma 6.3 (the continuum embedding of the inductive limit). *Let $(\beta_M)_M$ be the model-sheet family of injective $*$ -homomorphisms $\beta_M : \mathfrak{W}_M \rightarrow \mathfrak{W}_{\text{ct}}$ with $\beta_N \alpha_M^N = \beta_M$ for $M \leq N$ (model sheet §2; the scaling equation $R_N^\infty R_M^N = R_M^\infty$). Then there is a unique injective $*$ -homomorphism $\beta : \mathfrak{W}_\infty \rightarrow \mathfrak{W}_{\text{ct}}$ with $\beta \alpha_M^\infty = \beta_M$ for every M — the standing datum (3.1) of the Weyl-reduction Corollary 3.1.*

Proof. An injective $*$ -homomorphism of C^* -algebras is isometric, so every β_M and every $\alpha_M^N, \alpha_M^\infty$ is isometric. On the dense subalgebra $\bigcup_M \alpha_M^\infty(\mathfrak{W}_M)$ set $\beta(\alpha_M^\infty(A)) := \beta_M(A)$. Well defined: if $\alpha_M^\infty(A) = \alpha_{M'}^\infty(A')$ with $M \leq M'$, then $\alpha_{M'}^\infty(\alpha_M^{M'}(A) - A') = 0$ and $\alpha_{M'}^\infty$ is isometric, so $A' = \alpha_M^{M'}(A)$ and $\beta_{M'}(A') = \beta_{M'} \alpha_M^{M'}(A) = \beta_M(A)$. It is a $*$ -homomorphism on the union and isometric there, hence extends uniquely to the completion $\mathfrak{W}_\infty$; the extension is an isometric $*$ -homomorphism, therefore injective, and satisfies $\beta \alpha_M^\infty = \beta_M$. Uniqueness holds by density. $\square$

Theorem 6.4 (full-sequence identified fixed-coarse theorem). *Set*

$$\chi_M(\xi) := \lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi)));$$

the limit exists for every M and ξ by Theorem 6.2. Then, by the Weyl-reduction document (its Theorem 1.1, Corollary 3.1, and Proposition 4.1, all proved there):

1. *for every M there is a unique state ρ_M on $\mathfrak{W}_M$ with $\rho_M(W_M(\xi)) = \chi_M(\xi)$, and $\omega_N(\alpha_M^N(A)) \rightarrow \rho_M(A)$ for every $A \in \mathfrak{W}_M$;*
2. *the family $(\rho_M)_M$ is projectively consistent, and there is a unique state Ω_∞ on the inductive-limit algebra $\mathfrak{W}_\infty$ with $\Omega_\infty \circ \alpha_M^\infty = \rho_M$;*
3. *$\Omega_\infty = \omega_{L,\lambda}^{\text{ct}} \circ \beta$ on $\mathfrak{W}_\infty$: the full-sequence fixed-coarse limit of the lattice $P(\varphi)_2$ ground states is the pullback of the continuum torus $P(\varphi)_2$ vacuum;*
4. *all these states are invariant under the $\mathbb{Z}_2$ field symmetry $\vartheta(W(\zeta)) = W(-\zeta)$: ϑ commutes with the linear embeddings α_M^N , and $\omega_N \circ \vartheta_N = \omega_N$ by simplicity of the lattice ground state and the $\mathbb{Z}_2$ symmetry of $K_{N,L}$ (model sheet §3), so the Weyl-reduction Proposition 4.1 applies.*

In particular the interacting fixed-coarse projective cluster state of the projective-compactness document is unique and identified: the cofinal-subnet statement is superseded by full-sequence convergence.

Proof. (1)–(3) are the conclusions of the Weyl-reduction Theorem 1.1 and Corollary 3.1. Their hypotheses are: existence of $\chi_M(\xi)$ for every M, ξ , and the identity $\chi_M(\xi) = \omega^{\text{ct}}(\beta_M(W_M(\xi)))$ — both supplied by Theorem 6.2 — together with Corollary 3.1’s standing datum (its display (3.1)), the injective $*$ -homomorphism β with $\beta\alpha_M^\infty = \beta_M$, which that document takes as given and which Lemma 6.3 supplies from the model-sheet family. For (4): α_M^N maps $W_M(\xi) \mapsto W_N(R_M^N \xi)$ with R_M^N linear, so $\vartheta_N \alpha_M^N = \alpha_M^N \vartheta_M$; $K_{N,L}$ commutes with the unitary Θ_N implementing $\varphi \mapsto -\varphi$ (the interaction is even), so by simplicity $\Theta_N \Omega_N = \pm \Omega_N$ and $\omega_N \circ \vartheta_N = \omega_N$; Proposition 4.1 of the Weyl-reduction document transfers the invariance to ρ_M and Ω_∞ . $\square$

7 Acceptance, ledger, and phase closure

Acceptance against the scoping statements

A6 is Theorem 4.1 (with rate); A7 is Theorem 5.2 (in the normal-ordered normalization of Remark 1.1(2)); A8 is Theorems 6.2 and 6.4. The V4 gate condition of the scoping phase table — “(II.11) closed; Weyl criterion invoked” — is met. The fresh adversarial pass of 16 August 2026 is also complete: it repaired the four-edge count, the real-part defect of the asymmetric mixed surrogate, the missing-input $j = 3$ derivation, and the $\lambda = 0$ compactness argument. Its independent verification record is `repair_2026_08_16/POST_REPAIR_AUDIT.md`.

claim	status	ground
Action prefactors: $ S_N - S_\infty \leq C_\xi 2^{-r\eta/2}$	proved	Lemma 3.1 (V1 eq. (18) + V2 filter tail)
Source phases: $\ e^{i\Phi_N} - e^{i\Phi}\ _{L^2} \leq C_\xi 2^{-r\eta/4}$	proved	Lemma 3.2 (coupling symbols)
Denominators bounded below; $\mathcal{E}_N \rightarrow \mathcal{E}_\infty$; $\mathcal{N}_N \rightarrow \mathcal{N}_\infty$	proved	Lemma 3.3 (V2 Thms 4.3, 7.2, Prop. 6.5)
A6 with rate $C_\xi(2^{-r\eta/4} + \varepsilon_N \log_N^{3/2})$	proved	Theorem 4.1
Free products $\tilde{Z}_N^{(0)} \rightarrow Z_0$, rate ε_N^2	proved	Lemma 5.1
$\tilde{Z}_N(t) \rightarrow Z_\infty(t)_\wedge$ all $t \geq t_*$ (with $t_* L \geq 1$); $\inf_N \tilde{E}_0(N) > -\infty$	proved	Theorem 5.2(a,b)
$\tilde{E}_j(N) \rightarrow \mu_j$; $E_1(N) - E_0(N) \rightarrow \gamma_L > 0$; $D_N(t) \rightarrow D_\infty(t)$	proved	Theorem 5.2(c) (scoping Lemma 3.2)
Spectral tail bound, both sides	proved	Lemma 6.1
Gate (II.11), pointwise in M, ξ	proved	Theorem 6.2 (scoping Lemma 3.1)
Full-sequence identified fixed-coarse theorem; $\mathbb{Z}_2$	proved	Theorem 6.4 (Weyl-reduction Thm 1.1, Cor 3.1, Prop 4.1)
V1 (14) integrand reading; import	erratum	Rem. 1.1(1) (proved in V3, Thm 3.1(ii))
(I1) restated on the harmonic reference	recorded	

claim	status	ground
V1 (15): spurious $-6c_{N,L}(2H\eta + H^2)$ removed (Appell)	erratum recorded	Rem. 1.1(1')
V2 eq. (9) decay exponent is -2 (transcribed -3 in two places)	erratum recorded	Rem. 1.1(1'')
Convention lock corrected: $(a, b) = (0, 0)$, no counterterm; propagated to V2 Cor. 5.2/Rem. 5.3, V3 §1, scoping A4(b)	breaking erratum recorded	Rem. 1.1(1''')
A7 normal-ordered normalization	amendment recorded	Rem. 1.1(2)
V2 Def. 1.1: edge column symmetrized; η_N real and of the periodic-lattice law	erratum recorded	Rem. 1.1(3)
Scoping A7 conclusions restated in the normal-ordered symbols	erratum recorded	Rem. 1.1(4)
Scoping Lemma 3.2: hypothesis (i) for $t \geq t_*$; density step corrected; Thm 5.2 restated for $t \geq t_*$	erratum recorded	Rem. 1.1(5)
Scoping A3(b): $\coth(t\gamma/2) - 1 \leq 2e^{-t_0\gamma}$ replaced	erratum recorded	Rem. 1.1(6)
V2 §§4,6 repaired: four-edge sector restored; real asymmetric-band surrogate; folded and exact-output symbols separated; $A_{\times,j}$ explicit; no unnamed tail	repair propa- gated	Rem. 1.1(7)
$P_r(\pm\pi 2^r) = 0$: the $D4$ mask kills the edge column	new lemma	Rem. 1.1(8); V2, remark after Def. 6.2
Standing hypothesis $tL \geq 1$ propagated; $t_0 = t_1 = t_* = \max(1, 1/L)$	erratum recorded	Rem. 1.1(9)
Scoping Lemma 3.2 conclusion restricted to $t \geq t_*$	breaking erratum recorded	Rem. 1.1(10)

claim	status	ground
Fresh adversarial pass over the assembled chain	complete: algebra, edge combi- natorics, symbol domains, periodic tails, finite-mode normaliza- tion, trace limits, and all propagated gate imports rechecked	16 Aug 2026 post-repair audit record

What was consumed

V1: Thm 5.4 with eqs. (12), (14), (15); Rem 5.5; Lem 7.1, eq. (18) (eq. (19) is context, not consumed). V2: Def 1.1, Prop 1.2, Lem 2.1, Def 2.2, Lem 3.2/eq. (9), eq. (13), Thm 4.3, eq. (18), Def 6.2, Lem 6.3, Prop 6.5, Thm 7.2. V3: Thm 3.1(ii),(v)–(vi) + Rem 3.2 (lattice dictionary), Thm 4.1, Thm 5.2, Thm 6.2, Thm 7.1, Cor 6.3, Prop 2.6(iii), Lem 6.1(iv), Rem 4.2, §1 (endpoint convention). Scoping note: Lemmas 3.1, 3.2 (proved there). Free-rate document: Lemmas 2.1, 3.1 (as repackaged in V2). Weyl-reduction document: Thm 1.1, Cor 3.1, Prop 4.1 (proved there). Model sheet: §2 ($D4$ embeddings), §3 (lattice operator, ground state, $\mathbb{Z}_2$). No external import is consumed in this document.

Phase V4 gate: A6, A7, A8 proved; (II.11) closed; Weyl criterion and identification invoked; erratum register updated. The independent adversarial pass over the assembled chain V0–V4 was completed on 16 August 2026; its findings and regression checks are recorded in `repair_2026_08_16/POST_REPAIR_AUDIT.md`. Thus Part II’s fixed-coarse program is complete at the stated import scope and under the standing parameter hypotheses.